

Notebook QA and Execution Report

Quality, execution and coverage verification for the Day 12 notebook package

Ruturaj Mokashi

QA purpose

This report summarizes the quality checks expected for the Day 12 notebooks and documentation package. The goal is stable local execution and professional submission hygiene.

Check area	Expectation	Status
Notebook execution	Code cells executed successfully and outputs saved.	Completed for delivered notebooks
Code style	Short cells, commented logic, no giant unreadable blocks.	Completed
Markdown style	Detailed HTML intro and output-aware conclusions.	Completed
Dependencies	No inline package installation.	Completed
Paths	No personal local paths embedded.	Completed
MVD coverage	Every MVD task mapped to notebook evidence.	Completed
Additional coverage	All missing topics closed in Notebook 5.	Completed
Charts	Charts extracted and reused in BI presentation.	Completed
PDF hygiene	Separate PDFs, consistent layout, references included.	Completed

Notebook inventory

Notebook	Main focus	Datasets	Core models
01 Linear Regression Advertising	Simple and multiple linear regression, OLS, residual diagnostics, statsmodels OLS.	Advertising	Linear Regression, OLS
02 Regularized Regression	Ridge, Lasso, ElasticNet, coefficient shrinkage, Lasso path, LassoCV.	Diabetes regression dataset	Ridge, Lasso, ElasticNet, LassoCV
03 GLM Count Models	Bike Sharing count prediction, Poisson, Negative Binomial, overdispersion.	Bike Sharing	Linear Regression, PoissonRegressor, Poisson GLM, Negative Binomial GLM
04 Logistic Regression	Binary classification, probability, thresholding, calibration and lift.	Customer campaign response	Logistic Regression
05 Advanced Diagnostics	Missing advanced topics and influence diagnostics.	Advertising, Bike Sharing, campaign response	OLS diagnostics, gradient descent, multinomial logistic regression

Risk notes

- The Advertising and Bike Sharing notebooks use safe fallback data where the exact Kaggle CSV is not present, while retaining local-file loading logic.
- Boston Housing was not used directly because modern scikit-learn removed it; a stable regression dataset was used for regularization.
- Charts and metrics should be regenerated if the local raw Kaggle files are replaced with different data versions.

References and source foundation

The documents use the completed Day 12 notebooks as the main project evidence. The theory explanations are also aligned with the uploaded course materials and official library documentation.

Source	How it was used
Uploaded course guides: leit03.pdf and Haupt_leitfaden03.pdf	OLS, least squares, residuals, regression line, correlation, interpolation/extrapolation.
Uploaded ML slides: Methode der Kleinsten Quadrate and Typen des Machinellen Lernens	Supervised learning, regression vs classification, gradient descent, logistic regression context.
scikit-learn Linear Models User Guide - https://scikit-learn.org/stable/modules/linear_model.html	Linear Regression, Ridge, Lasso, ElasticNet, Logistic Regression, PoissonRegressor.
scikit-learn Model Evaluation User Guide - https://scikit-learn.org/stable/modules/model_evaluation.html	MSE, MAE, R2, ROC-AUC, PR-AUC, precision, recall, F1, calibration.
statsmodels GLM documentation - https://www.statsmodels.org/stable/glm.html	Generalized Linear Models, Poisson family and model interpretation.
statsmodels NegativeBinomial documentation - https://www.statsmodels.org/stable/generated/statsmodels.discrete.discrete_model.NegativeBinomial.html	Negative Binomial implementation details and count-model context.
UCLA IDRE Negative Binomial Regression notes - https://stats.oarc.ucla.edu/r/dae/negative-binomial-regression/	Practical explanation of Negative Binomial as a model for overdispersed count outcomes.